Knowledge Representation and Learning

15. project proposals

se faccio il progetto in una sessione successiva allo scritto mi devo inscrivere su uniweb

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Bisogna aver passato lo scritto

slide + demo

→ mostrare il progetto on-line

→ esame singolo

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grade
- 50% scritto
- 50% project

- 10 giorni l'esame scritto
- non sono vincolata alla sessione in cui passo lo scritto

Minesweeper

Applicare qualche forma di logical reasoning

Minesweeper is a game where mines are hidden in a grid of squares. The objective is to discover all the mines by opening the cells. At each iteration you have to open a cell if the cell contains a mine you have lost, otherwise you will discover the number of mines in the surrounding cells, which can be used to decide the next cell to be opened.

- Encode any game state (as that shown in picture) in a set of formulas S and use a SAT to decide for every cell i, j if:
- i, j is safe i.e., $S \models \neg mine_{ij}$.
- i, j is unsafe i.e., $S \models mine_{ij}$.
- If you cannot decide if i, j is safe or not (i.e., $S \not\models \neg mine_{ij}$ and $S \not\models mine_{ij}$) use (weighted) model counting to decide the move that minimize the probebility of selecting a mine.

qui ci sono la mina

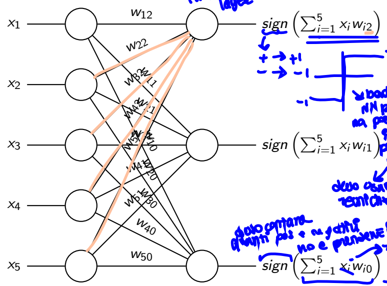
- casi in cui capisco dove è la mina in base a dove non è
- casi in cui non sono in grado di inferire nulla
  ↳ faccio questo usando il model counting (probabilistic)

Fitting binarized neural networks → implementa binari funct.

NN dove pesi e act funct sono binary (1, −1)

le Bin NN sono + trasparenti proprio xk usano bin funct

Binarized neural networks are nets in which both the weights and activations are binary. Their performance can be surprisingly good, and their implementation can be extremely efficient. In this project you have to show how to fit a binary neural network using SAT. Consider a one layer binary network with *n* neurons (n=5 in the picture).



+ → +1
− → −1

+1

−1

bad perk le NN binarie non posso usare il gradient per ottimizzare le bin NN → devo usare altre tecniche

devo contare quanti pos e negativi ho e prendere il max → of exclusive or → lo vedo come

where the unknown model parameters w_i are also binary and the function $sign(x)$ returns -1 or 1 based on the sign of x. Given a training set of data/label pairs $\{\mathbf{x}^{(i)}, \mathbf{y}^{(i)}\}_{i=1}^{d}$, where each $\mathbf{x}^{(i)} = (x_1^{(i)}, \ldots, x_n^{(i)}) \in \{-1, 1\}^n$, and $\mathbf{y}^{(i)} \in \{-1, 1\}^m$, one has to choose the model parameters w_{ij}, in order to maximize the correct predictions of the net.

Stesso con Noi
Essendo che tutto è booleano può essere tradotto in prop. logic

5 somma el.
· 3 posit → 1

Tutto da codificare con prop logic

neurone → prop. var → 1 True, → −1 False

weights → prop. var → 1 True, → −1 False

Come fare il learning?

Scegliere i parametri che fittano meglio sul training data

training data = {−1, 1, 1, −1, 1} input → out {1, −1, 1}

3 prop formula φ_1 φ_2 φ_3 input training

Fitting binarized neural networks

esempi

pero usare il maxsat per fissare il numero di training data che ho in modo soddisfacente

1. write a method that generates the encoding in max-sat of a binarized neural network layer with m inputs and n outputs
2. Use the previous method to generate the encoding in max-sat of a binarized neural network that stacks a set of binarized layers.
3. Define a method for train a specified binarized neural nework using max-sat
   - binary step function
   - Relu, Sigmoid, Maxsat → non linear function
4. Generate train and test data for at least 3 binary functions with 5 inputs and 5 outputs and evaluate the performance with different network configurations

for more details on binarized neural network see e.g. [MM08].

Weighted model counting via Knowledge compilation

1. Write a method that transforms a formula in sd-DNNF form, and use this method for computing the weighted model count of the formula.
2. Check the correctness of your algorithm by comparing the results of your method with the explicit computation of weighted model counting via truth table.
3. write a method estimates #SAT using the `sampleSat` algorithm, and compare the result of the approximated counting with the result obtained by the exact counting.

Probabilistic reasoning via WMC (se in coppia quello di prima + questo)

- Implement probabilistic inference in bayesian netwoirk using weighted model counting as explained in class.
- Use some dataset availeble in the python package `bnlearn` to test your implementation

Planning ~~with~~ (Max)Sat No!

Planning domain

- Let $\mathcal{P} = \{p_1, \ldots, p_n\}$ a set of propositional variable
- Any set of propositional variables $s \subset \mathcal{P}$ is a state
- An action $a = (pre(a), eff^+(a), eff^-(a))$
    - $pre(a)$ is a formula in $\mathcal{P}$, the precondition of a
    - $eff^+(a) \subseteq \mathcal{P}$ the positive effects of a
    - $eff^-(a) \subseteq \mathcal{P}$ the negative effects of a
- if $s \models pre(a)$, then $a(s) = s \cup eff^+(a) \setminus eff^-(a)$

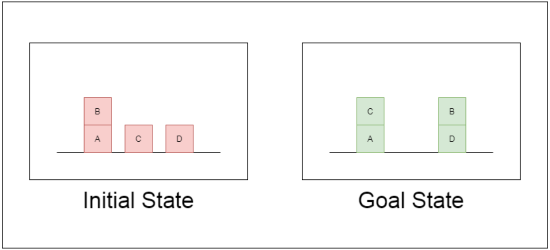
Planning problem

Given a set of actions A, an initial state s_0 and a goal g, which is a formula in $\mathcal{P}$, find a plan, i.e., a sequence of actions $a_1, a_2, \ldots, a_k$ such that

$$a_k(a_{k-1} \ldots a_2(a_1(s)), \ldots) \models g \tag{1}$$

Planning with (Max)Sat

- codify the problem of finding a plan of length less then or equal to k for reaching the goal g from an initial state s_0 as a SAT problem.
- Create a planning domain to solve the planning problem shown in the following picture:



- you can move blocks only if they don't have other blocks on top,
- you can mode them either on top of other blocks or on the table.
- find the smallest k for which a plan exists.

Grounding First Order Logic

Grounding

The grounding of a first order formula Φ that contains no constant anf function symbols on a domain with n elements is recursively defined as follows:

$$
\begin{aligned}
Ground(\forall x\, \phi, A) &= \bigwedge_{a \in A} Ground(\phi[s/a], A) \\
Ground(\exists x\, \phi, A) &= \bigvee_{a \in A} Ground(\phi[x/a], A) \\
Ground(\phi \circ \psi, A) &= Ground(\phi, A) \circ Ground(\psi, A) \\
Ground(\neg \phi, A) &= \neg Ground(\phi, A)
\end{aligned}
$$

Grounding First Order Logic

- Implement a system that ground first order sentence Φ on a finite domain $\{a_1, \ldots, a_n\}$;
- Use a SAT solver to check satisfiability of the grouned formula;
- If the grounding of Φ is satisfiable, extract from the truth assignment a first order interpretation on the domain $\{a_1, \ldots, a_n\}$ that satisfies Φ.
- use a large language model to translate a set of sentences in FOL formulas. By incrementally ground them with 1, 2, . . . , n constants search for a model for the sentences.

From NLP to FOL

- The goal of this project is to develop a bot model that translates natural-language (NL) statements into first-order logic (FOL) formulas.
- The recommended language nodel to be fine-tuned is the LLaMA-7B model: `https://huggingface.co/docs/transformers/main/model_doc/llama`
- The dataset to be used is the MALLS dataset: (https://huggingface.co/datasets/yuan-yang/MALLS-v0)

Reference

reference per la Bin NN

Marc Mezard and Thierry Mora. *Constraint satisfaction problems and neural networks: a statistical physics perspective.* 2008.